Dim3: A 3D Object Understanding Machine

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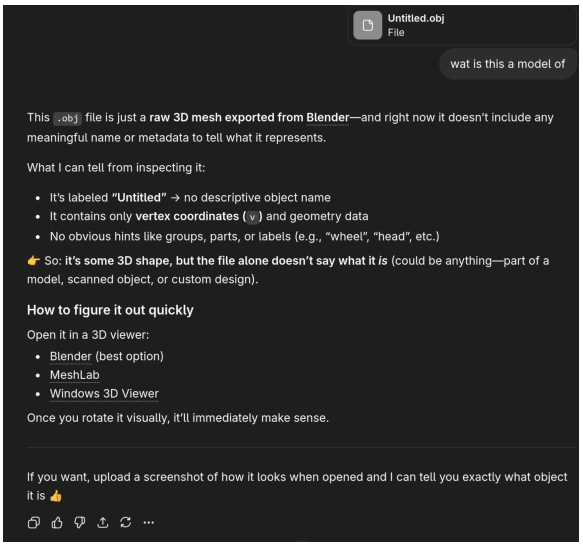
Problem Specification

Large Language Models have transformed and expanded tremendously throughout the recent years. At one point they were confined to text, but now they've progressed and gained the ability to process images, audio, and video footage. Despite all the progression, LLMs still can't process 3D objects. Millions of 3D object files in formats like .obj and .stl encode rich spatial and structural information that current LLMs cannot interpret and identify directly. As seen in the image, even the most advanced LLMs can't parse and use the content of an .obj file, responding with only that it contains vertex coordinates. They rely on the description in the metadata to classify. A way to bridge this gap is to have a preprocessing layer that is able to translate raw 3D geometry into natural language that an LLM can understand.



This project introduces *Dim3*, a Query Answering Machine designed as a proof-of-concept for translating 3D objects into meaningful textual insights. The system accepts any 3D file as input and converts it into a standardized .obj format to ensure consistent data representation. Using this unified structure, Dim3 analyzes the object and answers informative questions (see appendix).

Our core research question is whether or not geometric features extracted from 3D mesh files can be used to accurately classify and describe 3D medical objects.

Project Idea

Dim3 is a classification model that takes any 3D file as input, converts it into an .obj file and produces a semantic description of the object. It is designed to operate as a preprocessing component of an LLM.

For this project we limit ourselves to medical objects, with plans to expand to all 3D objects. The system targets different medical and anatomical categories like bones, organs, prosthetics, surgical tools, all sourced from Thingiverse's pages that can be tagged. This domain was chosen because it's important to correctly classify medical 3D models in clinical/educational context as misclassification can have serious consequences. The interface of our system will be a Python web app that allows users to upload a 3D file and get a description of it ready for LLM processing.

Data Set

We will build a custom dataset using tag-based scraping on Thingiverse. We'll make a web crawler that goes through different Thingiverse pages, each page with a different medical category as a tag. The crawler will collect their download links and retrieve files along with their labels to serve as truth. All files will be converted to .obj. We will have a target of 200-400 models per class (1000-2000 total).

Limitations & Risks

- Label Noise: Creator-assigned tags on Thingiverse can be ambiguous or inaccurate, this can be mitigated by manual checking of a sample.
- Class Imbalance: Bones are more represented than surgical tools on Thingiverse. Stratified sampling during data collection mitigates this.

- Scraping Constraints: Rate limits/CAPTCHAs can restrict access

Appendix:

Dim3 will provide answers to the following questions:

- What is this object (in our model what class does it belong to)?
- How many vertices, edges, and faces does the model have?
- What kind of symmetries does it have?
- What is the volume and surface area of the object?
- What are the primary geometric features (e.g., flat surfaces, curves, edges)?
- Where is its center of mass?
- What is the bounding box or spatial dimensions (height, width, depth)?
- Does the object contain repeating patterns or structures?
- What is the level of geometric complexity (simple vs. intricate)?
- Is the object manifold and watertight?
- Are there any disconnected components or parts?